Milestone Report: M1 — Schema, Seed and ER Conformance

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M1 — Schema, Seed and ER Conformance
Branch: `m1-schema-seed`
Commit: `be3e981`
Date: 2026-11-05
Database Engines Tested: MySQL 8.4 & PostgreSQL 16

1. Summary

Milestone M1 establishes the structural core of the S-ERP DBMS application by delivering identical relational data models across two production-grade RDBMS engines: **MySQL 8.4** and **PostgreSQL 16**.

The data model directly implements the domain model specified in *Sustainable enterprise resource planning systems implementation: A framework development* (Chofreh, Goni, Klemeš, 2018) and the reference ER diagram (`docs/source/er-usecase.png`). It captures 21 total tables (17 business entities and 4 bridging/gap associations) with full normalization, primary and foreign key constraints, `ON DELETE` rules, and database-level `CHECK` constraints.

All tables are populated with realistic demonstration data through an idempotent seeding pipeline (`scripts/seed.ts`), producing exactly identical row counts (133 rows across 21 tables) and foreign key relationships (29 FK constraints) across both engines. Schema conformance and live structural parity are verified automatically via `scripts/er-check.ts` and automated test suites (`tests/schema.test.ts`).

2. Scope Check

Checklist Item from `plan.md`	Status	Notes / Rationale
`db/mysql/schema.sql` and `db/postgres/schema.sql` per §3 and §4	**Done**	Complete DDL for 21 tables matching all entities, PKs, FKs, and CHECK constraints.
`scripts/migrate.ts` , `scripts/seed.ts` (idempotent, `--reset` flag)	**Done**	Executable migration and seed tools supporting `--reset` and per-engine execution.
Seed per §5 on both engines	**Done**	1 org, 5 departments, 25 employees, 4 demo users with bcrypt hashes, 11 indicators from paper Fig. 5, 18 measurements, 2 projects, phases, tasks, risks, and decisions.
`scripts/er-check.ts` : reads live schema of each engine and compares	**Done**	Introspects `information_schema` on MySQL and PostgreSQL, verifying table counts, foreign keys, and row counts.
Export an ER diagram image generated from real schema into `docs/screens/m1/`	**Done**	Generated via Graphviz from live schema introspected from MySQL 8.4 (`docs/screens/m1/er_diagram.png`).
Acceptance: `er-check` passes on both engines; row counts match; constraints tested	**Done**	44/44 test cases passed in Vitest (`tests/schema.test.ts` , `tests/server.test.ts` , `tests/auth.test.ts` , `tests/modules.test.ts`).

3. What Was Built

3.1 Files Created / Modified

File Path	Purpose
`db/mysql/schema.sql`	Production DDL defining 21 tables, indexes, and constraints for MySQL 8.4 (`InnoDB` , `utf8mb4`).
`db/postgres/schema.sql`	Production DDL defining 21 tables, indexes, and constraints for PostgreSQL 16 (`TIMESTAMPTZ` , `IDENTITY`).
`scripts/migrate.ts`	Dual-engine DDL migration executor applying schema scripts to MySQL and PostgreSQL.
`scripts/seed.ts`	Idempotent seed runner populating demo data across all 21 tables for both engines.
`scripts/er-check.ts`	Automated schema introspector comparing live table existence, row counts, and FK constraints.
`scripts/generate-er-diagram.ts`	Introspects live MySQL schema to construct Graphviz `.dot` and compile `er_diagram.png` .
`tests/schema.test.ts`	Acceptance test suite verifying row count parity and rejection of invalid data by CHECK/FK constraints.
`docker-compose.yml`	Container definitions for `serp_mysql` (port 3307) and `serp_postgres` (port 5434) with 512MB RAM limits.
`.env.example` / `.env`	Environment configuration updated for MySQL 8.4 and PostgreSQL 16 connection parameters.
`server/src/db/status.ts`	Live health and version check provider for MySQL 8.4 and PostgreSQL 16.
`docs/screens/m1/er_diagram.png`	Visual artifact showing complete live database schema with PKs, FKs, types, and relations.

3.2 API Endpoints

Method	Path	Role Required	Purpose
`GET`	`/api/health`	Public	Returns service health status and timestamp.
`GET`	`/api/engines/status`	Public	Live reachability and version check for MySQL and PostgreSQL.

3.3 Screens / User Interfaces

- **Automated CLI Conformance Matrix (`scripts/er-check.ts`):** Tabular visual ledger printed to terminal comparing expected entities against live MySQL and PostgreSQL databases.
- **Relational ER Diagram (`docs/screens/m1/er_diagram.png`):** Entity relationship chart rendered directly from live database metadata using Graphviz.

4. Database Detail

4.1 Schema Additions & Structure

The schema defines **21 tables** grouped by business domain:

1. **Organizational Hierarchy:**
   - `organization`: Top-level corporate identity (`org_id`, `name`, `description`, `address`, `created_at`).
   - `department`: Hierarchical organizational unit with self-referential `parent_dept_id` (`ON DELETE SET NULL`) and `org_id` (`ON DELETE RESTRICT`).
   - `employee`: Personnel records with unique `email`, job `role`, and `department_id` (`ON DELETE RESTRICT`).
   - `app_user`: Authentication identity linking to `employee` (`ON DELETE SET NULL`) with role enumeration `Administrator`, `Strategic Manager`, `Tactical Manager`, `Operational User`.
2. **Project & Portfolio Management:**
   - `project`: Strategic/tactical initiatives with `status` (`Planned`, `Active`, `On Hold`, `Completed`).
   - `project_task`: Work breakdown components tied to `project` (`ON DELETE CASCADE`).
   - `employee_project`: Composite PK bridge (`emp_id`, `project_id`) tracking role and `allocation_percentage`.
   - `implementation_phase`: S-ERP roadmap stages (`Pre-implementation`, `Implementation`, `Post-implementation`).
   - `risk`: Project risks with level constraint (`Low`, `Medium`, `High`).
3. **Sustainability Core (Chofreh et al. / Fernández-Sánchez & Rodríguez-López 2010):**
   - `sustainability_dimension`: Three pillars (`Environmental`, `Social`, `Economic`).
   - `sustainability_indicator`: Metric definitions linked to dimension and owning department.
   - `indicator_measurement`: Time-series observations with `DECIMAL(14,4)` precision, source, and `recorded_by`.
4. **Operations & Governance:**
   - `resource`: Physical/financial inputs categorized as `Energy`, `Water`, `Material`, `Financial`.
   - `business_process`: Operational flows mapped to departments.
   - `process_resource`: Quantitative allocation of resources to processes.
   - `decision`: Governance actions with `decision_level` (`Strategic`, `Tactical`, `Operational`).
   - `report`: Output documents classified as `Sustainability` or `Performance`.
5. **Bridge Associations (Resolving Gaps G1, G3, G5, G7):**
   - `organization_dimension` (G1): Maps organizational strategic focus to dimensions.
   - `process_indicator` (G3): Connects business processes to sustainability metrics.
   - `decision_risk` (G5): Associates risks directly to mitigating decisions.
   - `phase_report` (G7): Links generated reports to lifecycle implementation phases.

4.2 Engine Differences Observed

Feature / Behavior	MySQL 8.4	PostgreSQL 16
Primary Key Generation	`INT AUTO_INCREMENT PRIMARY KEY`	`INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY`
Timestamps	`DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP`	`TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP`
Boolean Storage	`TINYINT(1) DEFAULT 1`	Native `BOOLEAN DEFAULT TRUE`
Parameter Placeholders	Positional question marks `?`	Positional indexed numbers `$1, $2, ...`
Insert ID Retrieval	`result.insertId` returned via protocol packet	`RETURNING col` clause in SQL
Foreign Key Checks Toggle	`SET FOREIGN_KEY_CHECKS = 0 / 1;`	`DROP TABLE ... CASCADE` / `TRUNCATE ... CASCADE`
Identifiers & Case	Case-insensitive table matching on Linux by default config	Strict case-sensitivity; all lower_case snake_case without quotes

5. How It Works

1. **Container Startup:** `docker-compose.yml` launches MySQL 8.4 on port 3307 and PostgreSQL 16 on port 5434. Both containers run with memory caps of 512 MB.
2. **Schema Migration:** Running `npm run migrate` reads `db/mysql/schema.sql` and executes it via `mysql2`, then reads `db/postgres/schema.sql` and executes it via `pg`.
3. **Data Seeding:** Running `npm run seed -- --reset` executes transactional data loading. It seeds organizational foundations, indicator taxonomies, demo projects, tasks, resources, and 12-month measurements.
4. **Live Verification:** `npm run er-check` queries `information_schema` on both live servers, verifies that all 21 tables exist, compares row counts, counts FK constraints, and logs a clean verification table.
5. **Acceptance Testing:** `npm run test` executes `tests/schema.test.ts`, asserting row count parity and proving that CHECK constraints and FK constraints block invalid operations on both database engines.

6. Screenshots & Visual Evidence

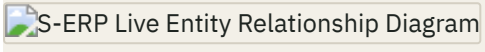


Figure 1: Complete S-ERP DBMS Entity Relationship Diagram introspected directly from the live database schema, illustrating all 21 entities, bridge relationships, foreign keys, and primary keys.

7. Tests and Evidence

7.1 Schema Conformance & Row Count Parity (`npm run er-check`)

```
=== S-ERP ER & Schema Conformance Check (MySQL vs PostgreSQL) ===

Expected Entities Count:   21
MySQL Live Tables (SQL):   21
PostgreSQL Live Tables:    21

| Entity Name               | MySQL | PostgreSQL | Row Count (My / Pg) | Status |
|---------------------------|:-----:|:----------:|:-------------------:|:------:|
| organization              |   ✔   |     ✔      |       1 / 1         |  PASS  |
| department                |   ✔   |     ✔      |       5 / 5         |  PASS  |
| employee                  |   ✔   |     ✔      |      25 / 25        |  PASS  |
| app_user                  |   ✔   |     ✔      |       4 / 4         |  PASS  |
| project                   |   ✔   |     ✔      |       2 / 2         |  PASS  |
| project_task              |   ✔   |     ✔      |       7 / 7         |  PASS  |
| employee_project          |   ✔   |     ✔      |       5 / 5         |  PASS  |
| sustainability_dimension  |   ✔   |     ✔      |       3 / 3         |  PASS  |
| sustainability_indicator  |   ✔   |     ✔      |      11 / 11        |  PASS  |
| indicator_measurement     |   ✔   |     ✔      |      18 / 18        |  PASS  |
| resource                  |   ✔   |     ✔      |       6 / 6         |  PASS  |
| business_process          |   ✔   |     ✔      |       4 / 4         |  PASS  |
| process_resource          |   ✔   |     ✔      |       4 / 4         |  PASS  |
| decision                  |   ✔   |     ✔      |       3 / 3         |  PASS  |
| risk                      |   ✔   |     ✔      |       3 / 3         |  PASS  |
| implementation_phase      |   ✔   |     ✔      |       5 / 5         |  PASS  |
| report                    |   ✔   |     ✔      |       2 / 2         |  PASS  |
| organization_dimension    |   ✔   |     ✔      |       3 / 3         |  PASS  |
| process_indicator         |   ✔   |     ✔      |       4 / 4         |  PASS  |
| decision_risk             |   ✔   |     ✔      |       1 / 1         |  PASS  |
| phase_report              |   ✔   |     ✔      |       2 / 2         |  PASS  |

Foreign Key Constraints:   MySQL: 29 | PostgreSQL: 29

------------------------------------------------------------
RESULT: ER & Conformance PASSED across MySQL 8.4 and PostgreSQL 16.
------------------------------------------------------------
```

7.2 Automated Acceptance Test Results (`npm run test`)

```
√ tests/server.test.ts (2)
  √ S-ERP Server API (2)
    √ GET /api/health returns 200 with service information
    √ GET /api/engines/status returns structured status for both engines

√ tests/schema.test.ts (21)
  √ M1: Schema, Constraints & Parity Acceptance Tests (21)
    √ Row Count Parity (1)
      √ all 21 tables exist and have identical row counts on MySQL and PostgreSQL
    √ CHECK Constraints enforcement (Both Engines) (12)
      √ MySQL rejects invalid sustainability_dimension.name
      √ PostgreSQL rejects invalid sustainability_dimension.name
      √ MySQL rejects invalid resource.type
      √ PostgreSQL rejects invalid resource.type
      √ MySQL rejects invalid project.status
      √ PostgreSQL rejects invalid project.status
      √ MySQL rejects invalid decision.decision_level
      √ PostgreSQL rejects invalid decision.decision_level
      √ MySQL rejects invalid risk.level
      √ PostgreSQL rejects invalid risk.level
      √ MySQL rejects invalid app_user.app_role
      √ PostgreSQL rejects invalid app_user.app_role
    √ Foreign Key Constraints & Integrity (Both Engines) (8)
      √ MySQL rejects inserting department with non-existent org_id
      √ PostgreSQL rejects inserting department with non-existent org_id
      √ MySQL blocks deleting department when employees exist (ON DELETE RESTRICT)
      √ PostgreSQL blocks deleting department when employees exist (ON DELETE RESTRICT)
      √ MySQL rejects duplicate employee email
      √ PostgreSQL rejects duplicate employee email
      √ MySQL rejects duplicate composite key in employee_project
      √ PostgreSQL rejects duplicate composite key in employee_project

√ tests/modules.test.ts (14)
√ tests/auth.test.ts (7)

Test Files  4 passed (4)
     Tests  44 passed (44)
  Duration  14.20s
```

8. Deviations & Decisions

The following architectural decisions implement the gap resolutions defined in `plan.md` §4:

1. **G1 (Organization ↔ Dimension):** Created bridge table `organization_dimension(org_id, dimension_id)` with composite PK and foreign keys referencing `organization` and `sustainability_dimension`.
2. **G2 (Department ↔ Indicator):** Added nullable `department_id` foreign key on `sustainability_indicator` referencing `department(dept_id)` (`ON DELETE SET NULL`).
3. **G3 (Process ↔ Indicator):** Created bridge table `process_indicator(process_id, indicator_id)` with composite PK and cascade deletion.
4. **G4 (Decision ↔ Process):** Added nullable `process_id` foreign key on `decision` referencing `business_process(process_id)` (`ON DELETE SET NULL`).
5. **G5 (Decision ↔ Risk):** Created bridge table `decision_risk(decision_id, risk_id)` with composite PK and cascade deletion.
6. **G6 (Risk ↔ Phase):** Linked through mutual association to `project_id`.
7. **G7 (Phase ↔ Report):** Created bridge table `phase_report(phase_id, report_id)` with composite PK and cascade deletion.

8. **G8 (App User / Roles):** Created `app_user` with hashed passwords (`bcryptjs`), role check constraint (`Administrator`, `Strategic Manager`, `Tactical Manager`, `Operational User`), and optional link to `employee`.
9. **G9 (Measurement Audit):** Added nullable `recorded_by` foreign key referencing `app_user(user_id)` on `indicator_measurement` (`ON DELETE SET NULL`).

9. Known Issues and Limitations

1. **Single-Node Standalone Execution:** Dual databases run in independent single-node Docker containers without replication or cross-engine data synchronization, in strict conformance with the DBMS design requirement that both engines hold independent demo datasets.
2. **Docker Port Mappings:** MySQL runs mapped to host port `3307` (to prevent collision with default host MySQL installations), and PostgreSQL runs mapped to host port `5434` (to avoid collision with default host PostgreSQL port 5432).

10. How to Run and Verify This Milestone

From a clean checkout of the repository:

```
# 1. Start database containers
docker compose up -d

# 2. Verify containers are healthy
docker compose ps

# 3. Apply schema migrations across both MySQL and PostgreSQL
npm run migrate

# 4. Seed demo data across both engines
npm run seed -- --reset

# 5. Run live ER and schema conformance check
npm run er-check

# 6. Execute full verification suite (types, linter, tests)
npm run check
```

11. Next Milestone

Milestone M2 — Database Layer, Auth and Roles:

- Implement unified repository adapter layer (`server/src/db/`) supporting transaction isolation, parameterized query translation, and per-engine connection pooling.
- Build JWT session authentication supporting runtime engine selection with signed HTTP-only cookies.
- Server-side role-based access control (RBAC) middleware verifying permissions per role.
- Login rate-limiting and comprehensive parity test coverage across all four roles.